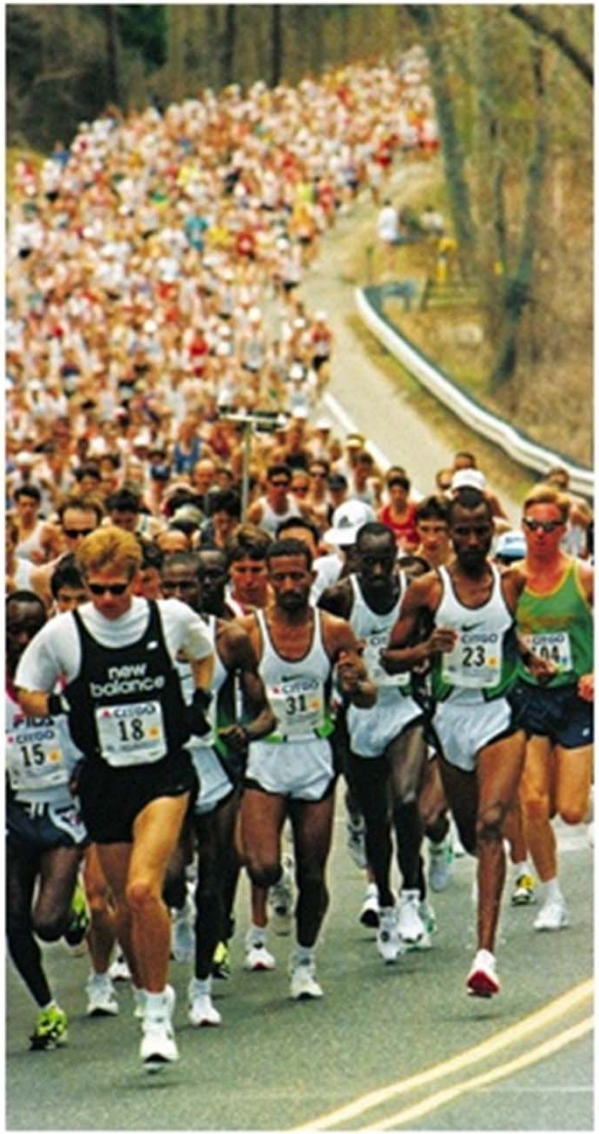
# Fatty acid oxidation



# Fatty Acid Catabolism

*Learning goals:*

- How fats are mobilized and transported in tissues
- How fats are oxidized
- How “ketone bodies” are produced

# Oxidation of Fatty Acids Is a Major Energy Source in Many Organisms

- About one-third of our energy needs comes from dietary triacylglycerols.
- About 80% of energy needs of mammalian heart and liver are met by oxidation of fatty acids.
- Many hibernating animals, such as grizzly bears, rely almost exclusively on fats as their source of energy.

# Hibernating Bears Get the Majority of Their Energy from Stored Fatty Acids



Stouffer Productions/Animals Animals

**Box 17-1**

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# Fats Provide Efficient Fuel Storage

- The advantage of fats over polysaccharides:
  - Fatty acids carry more energy per carbon because they are more reduced.
  - Fatty acids complex or carry less water because they are nonpolar.
- Glucose and glycogen are for short-term energy needs and quick delivery.
- Fats are for long-term (months) energy needs, good storage, and slow delivery.

# Hormones Trigger Mobilization of Stored Triacylglycerols

Lipases are activated by hormones glucagon and epinepherine.

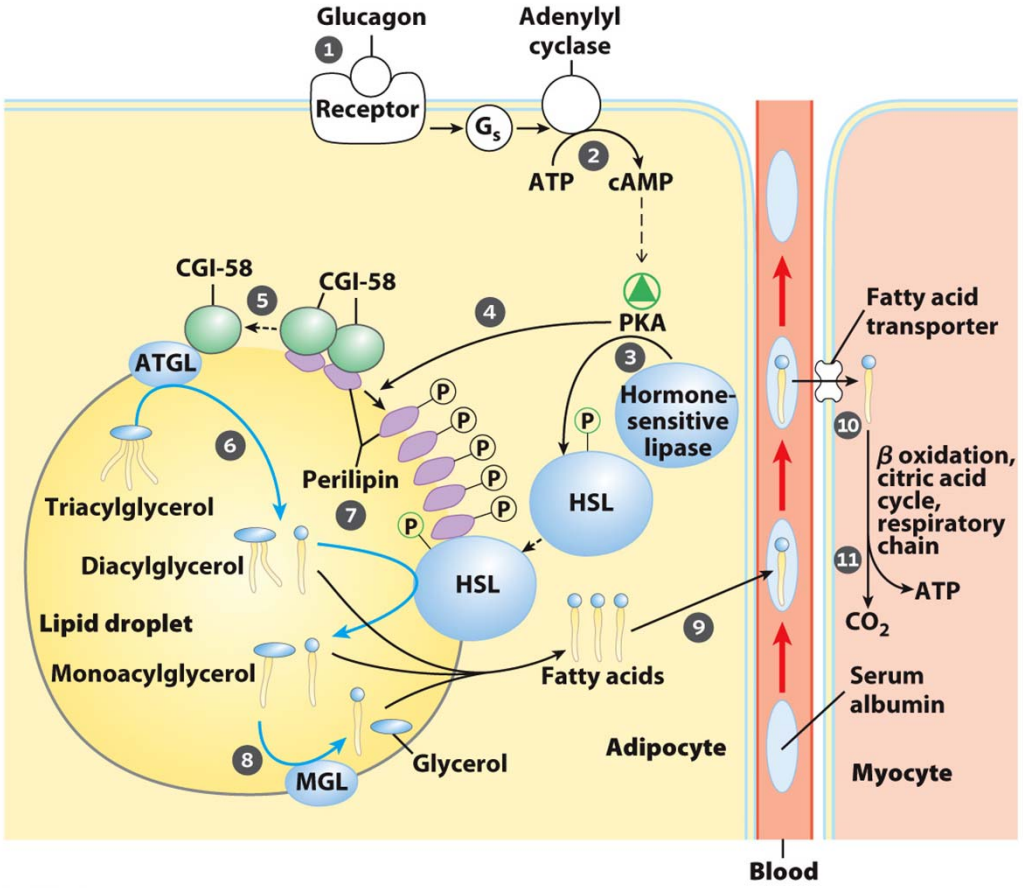



Figure 17-3
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# Transport of Fatty acids to the Mitochondria Requires Conversion to Fatty Acyl-CoA

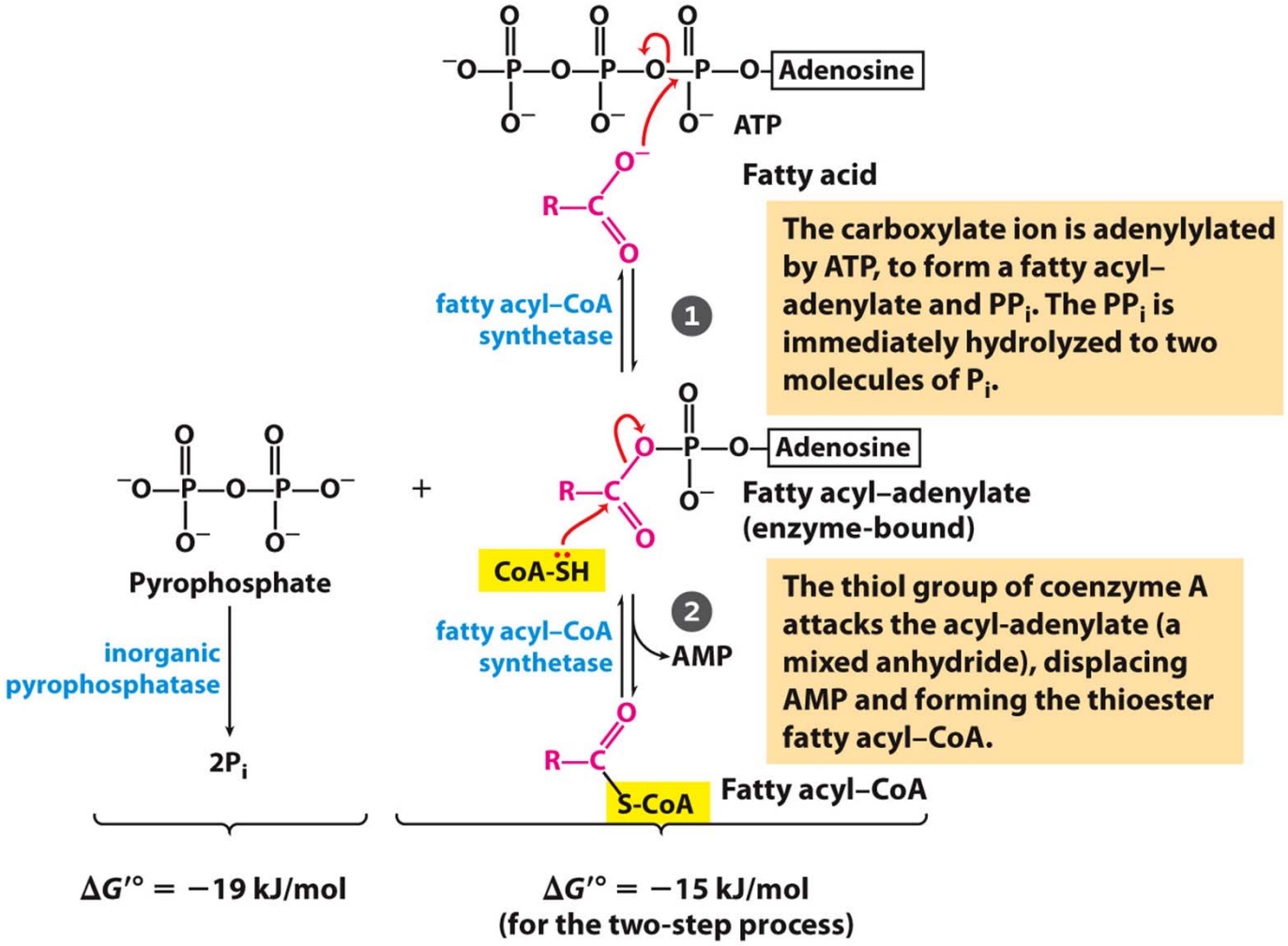
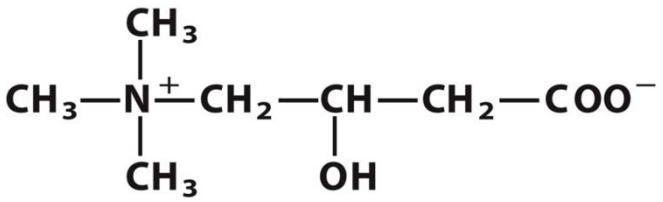



**Figure 17-5**
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# Fatty Acid Transport into Mitochondria

- Fats are degraded into fatty acids and glycerol in the cytoplasm of adipocytes.
- Fatty acids are transported to other tissues for fuel through the blood.
- $\beta$ oxidation of fatty acids occurs in mitochondria.
- Small (< 12 carbons) fatty acids diffuse freely across mitochondrial membranes.
- Larger fatty acids (most free fatty acids) are transported via acyl-carnitine/carnitine transporter.




**Carnitine**

Unnumbered 17 p653
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# Acyl-Carnitine/Carnitine Transport

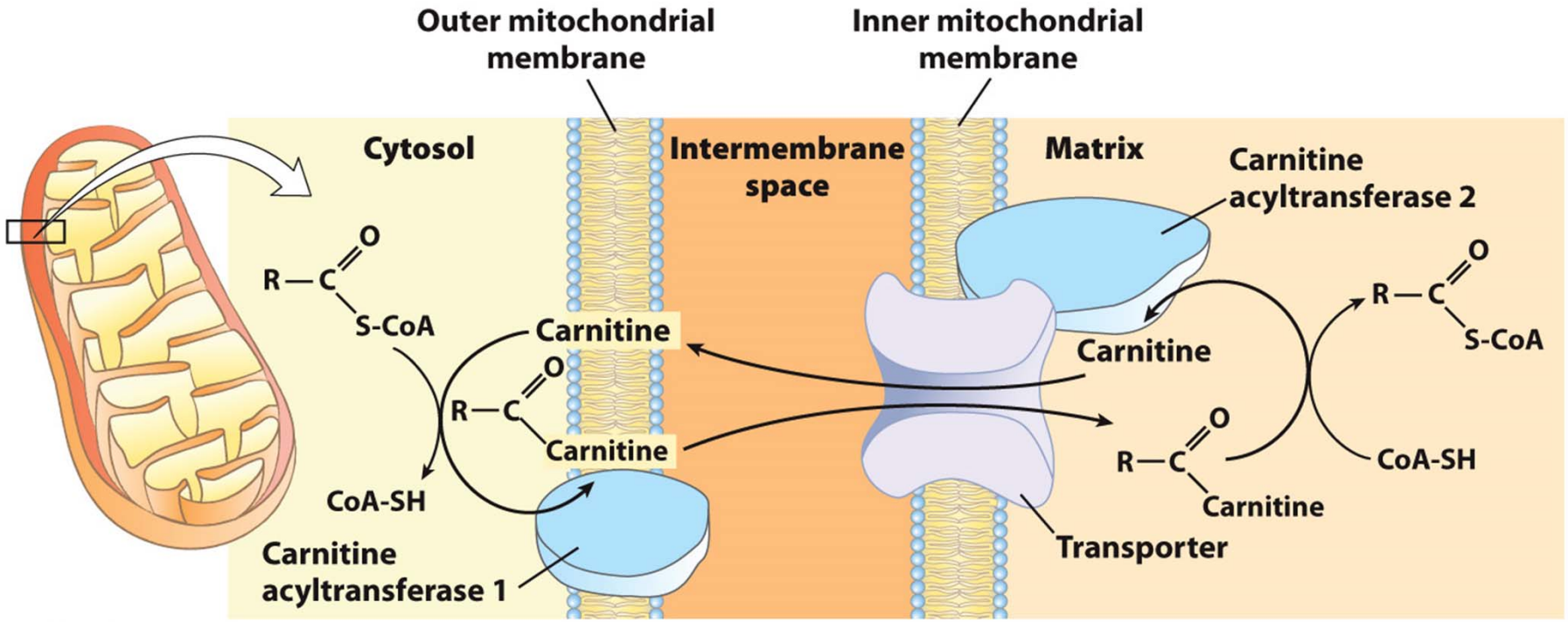



**Figure 17-6**
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# Fatty Acid Oxidation Occurs in the Mitochondria in Three Stages

- Stage 1 consists of oxidative conversion of two-carbon units into acetyl-CoA via $\beta$ oxidation with concomitant generation of NADH and $FADH_2$.
  - involves oxidation of $\beta$ carbon to thioester of fatty acyl-CoA
- Stage 2 involves oxidation of acetyl-CoA into $CO_2$ via citric acid cycle with concomitant generation NADH and $FADH_2$.
- Stage 3 generates ATP from NADH and $FADH_2$ via the respiratory chain.

# Stages of Fatty Acid Oxidation

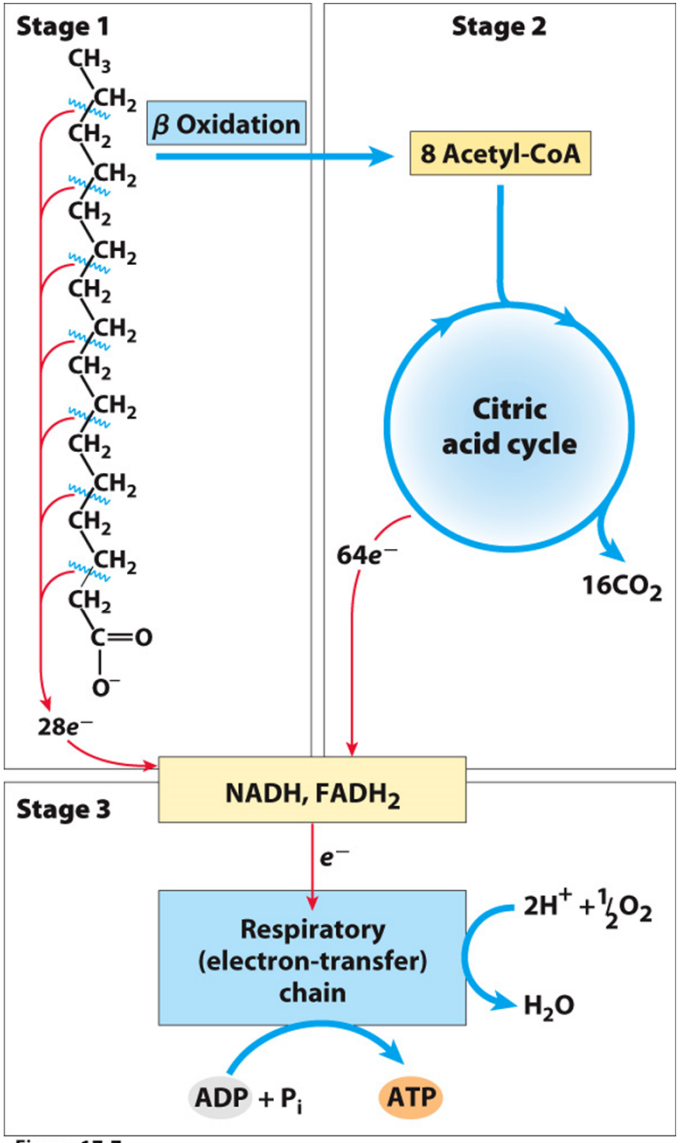
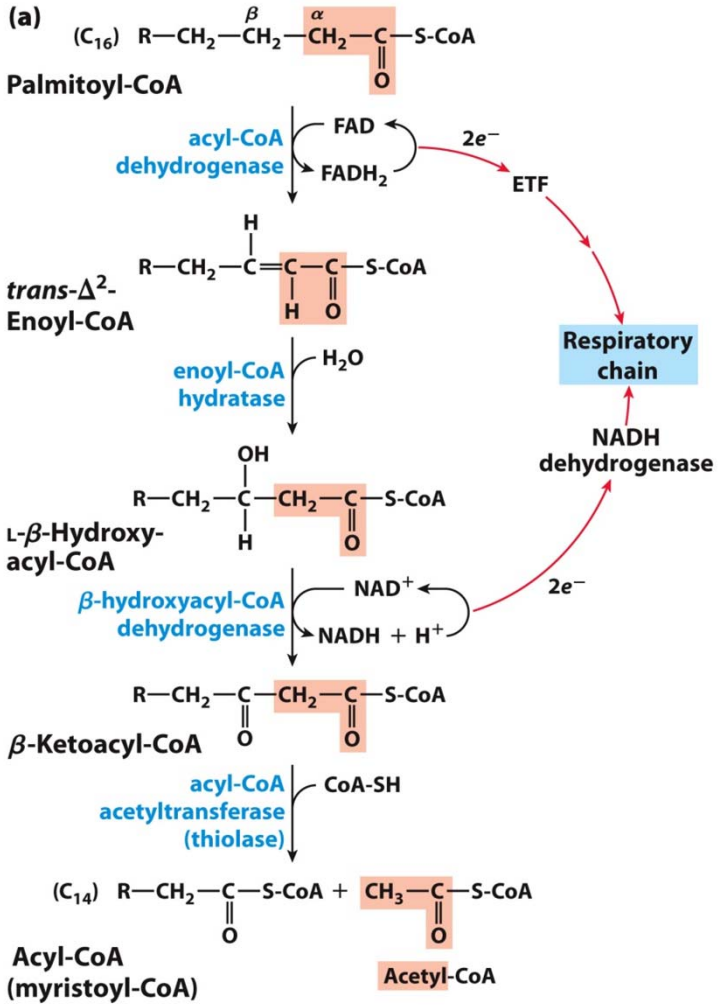



**Figure 17-7**
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# The β-Oxidation Pathway

Each pass removes one acetyl moiety in the form of acetyl-CoA.



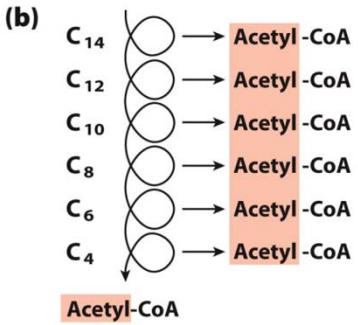




**Figure 17-8**
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# Each Round Produces an Acetyl-CoA and Shortens the Chain by Two Carbons

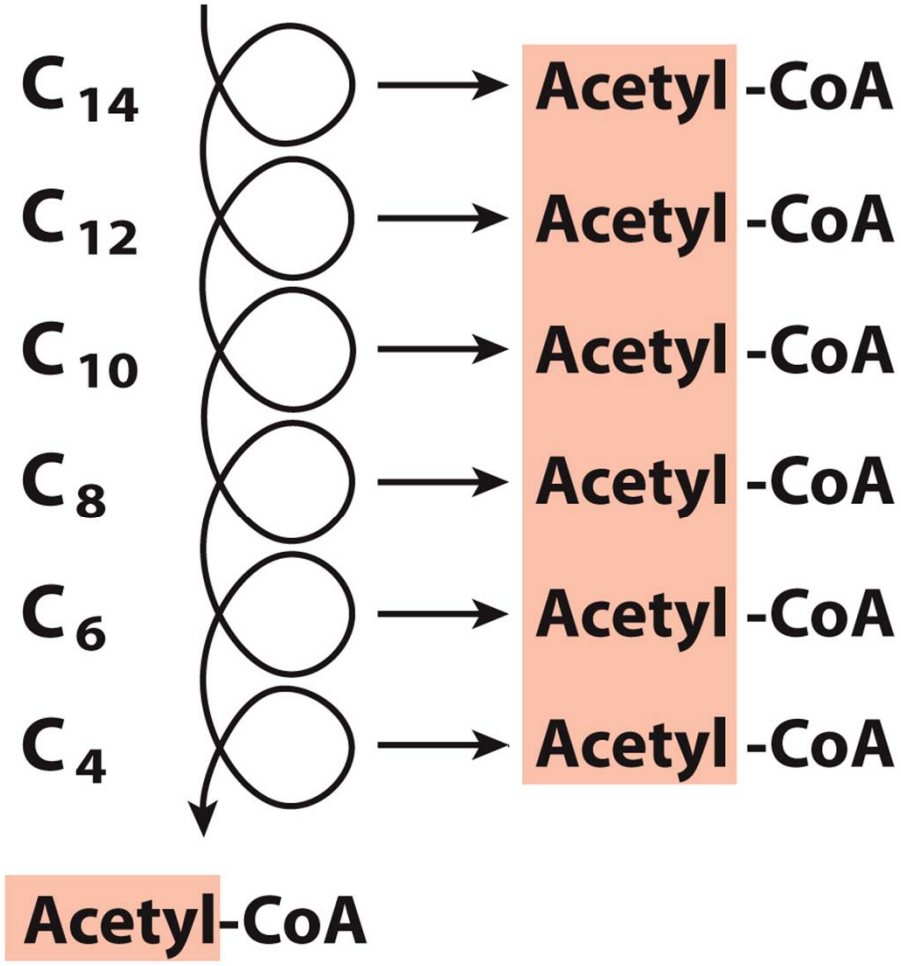



**Figure 17-8b**
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# Fatty Acid Catabolism for Energy

- For palmitic acid ($C_{16}$)
  - Repeating the previous four-step process six more times (seven total) results in eight molecules of acetyl-CoA.
    - $FADH_2$ is formed in each cycle (seven total).
    - NADH is formed in each cycle (seven total).

- Acetyl-CoA enters citric acid cycle and further oxidizes into $CO_2$.
  - This makes more GTP, NADH, and $FADH_2$.

- Electrons from all $FADH_2$ and NADH enter ETF.

# NADH and $FADH_2$ Serve as Sources of ATP

**TABLE 17-1** Yield of ATP during Oxidation of One Molecule of Palmitoyl-CoA to $CO_2$ and $H_2O$

| Enzyme catalyzing the oxidation step | Number of NADH or $FADH_2$ formed | Number of ATP ultimately formed[a] |
|---|---|---|
| **β Oxidation** | | |
| Acyl-CoA dehydrogenase | 7 $FADH_2$ | 10.5 |
| β-Hydroxyacyl-CoA dehydrogenase | 7 NADH | 17.5 |
| **Citric acid cycle** | | |
| Isocitrate dehydrogenase | 8 NADH | 20 |
| α-Ketoglutarate dehydrogenase | 8 NADH | 20 |
| Succinyl-CoA synthetase | | 8[b] |
| Succinate dehydrogenase | 8 $FADH_2$ | 12 |
| Malate dehydrogenase | 8 NADH | 20 |
| Total | | 108 |

[a]These calculations assume that mitochondrial oxidative phosphorylation produces 1.5 ATP per $FADH_2$ oxidized and 2.5 ATP per NADH oxidized.
[b]GTP produced directly in this step yields ATP in the reaction catalyzed by nucleoside diphosphate kinase (p. 516).

# Similar Mechanisms Introduce Carbonyls in Other Metabolic Pathways

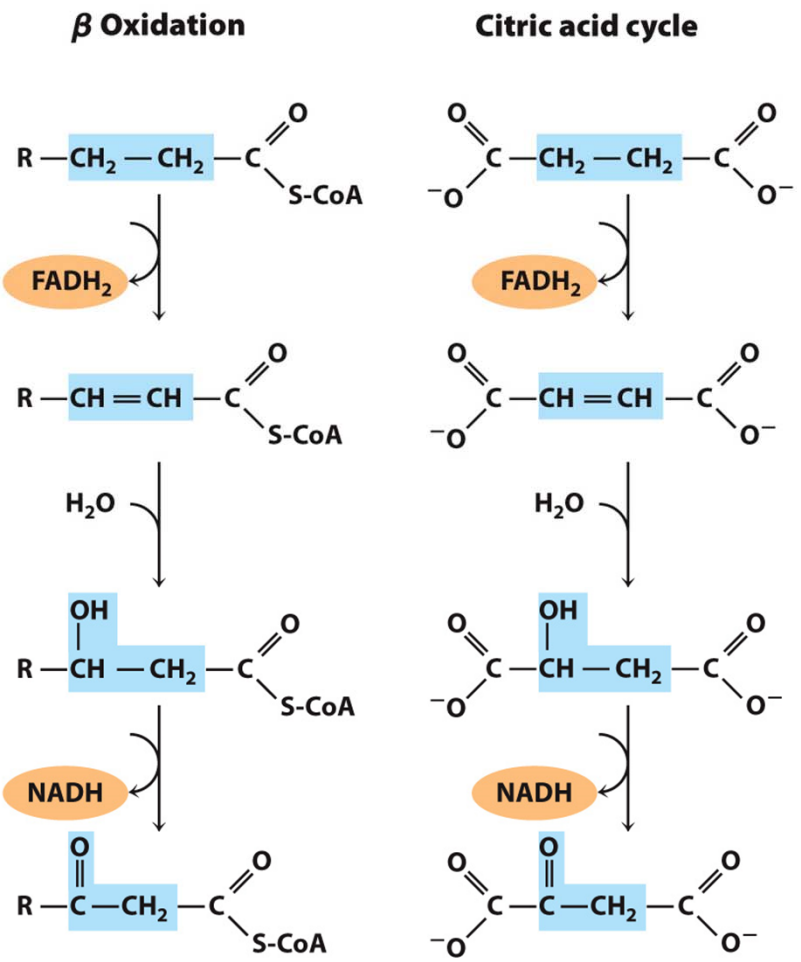



**Figure 17-9**
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# Coordinated regulation of fatty acid synthesis and breakdown

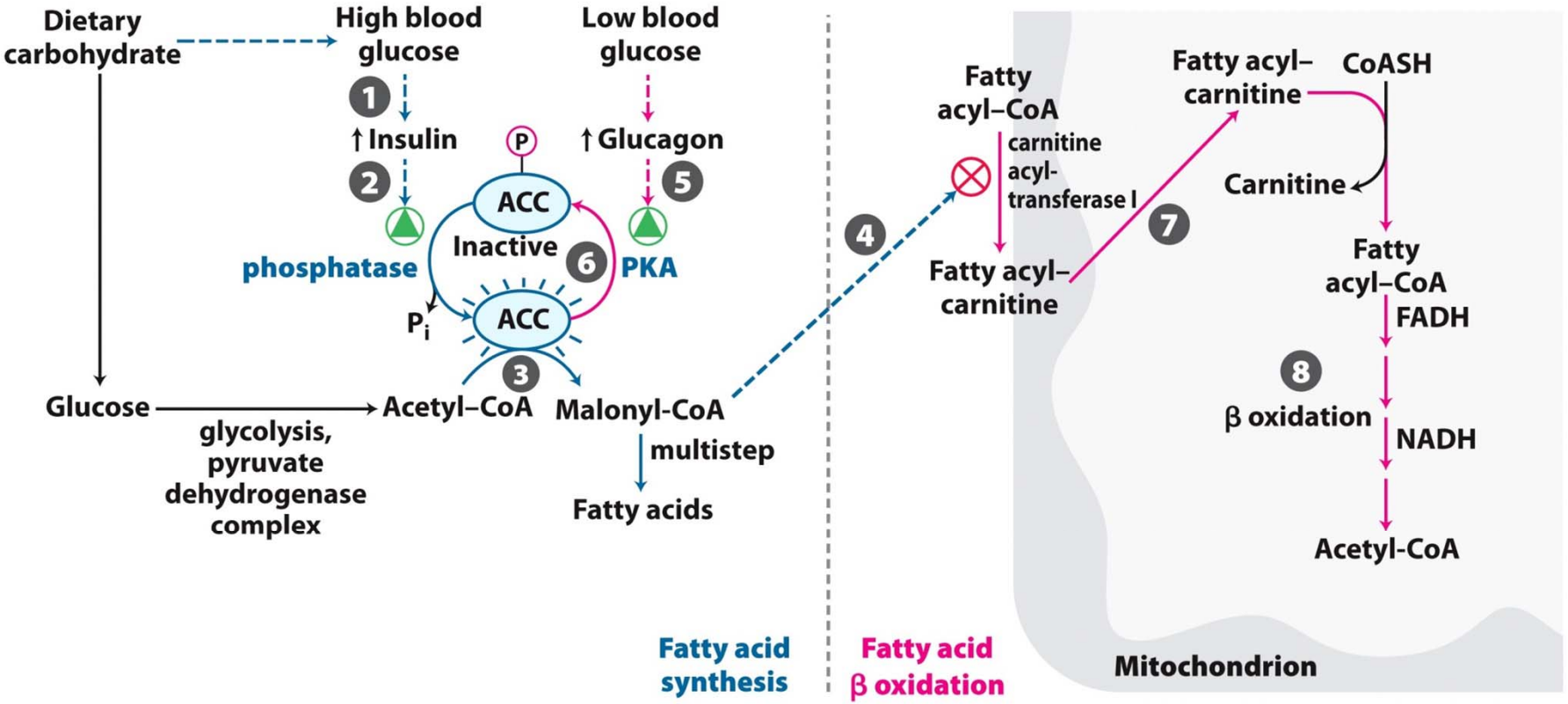
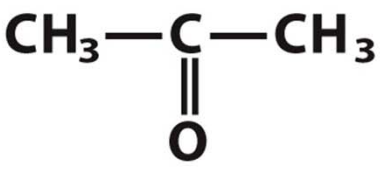


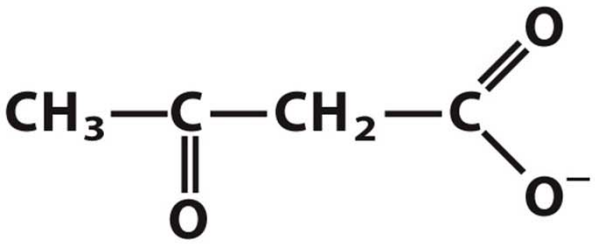

**Figure 17-13**
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# Ketone Bodies

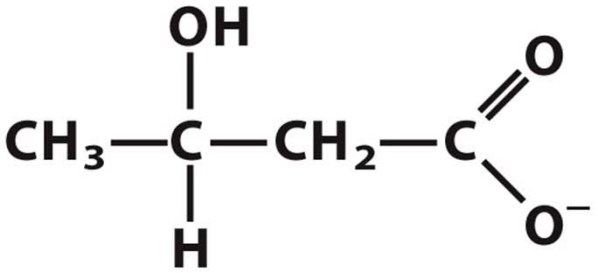
- Entry of acetyl-CoA into citric acid cycle requires oxaloacetate.
- When oxaloacetate is depleted, acetyl-CoA is converted into ketone bodies.
  - frees coenzyme A for continued $\beta$ oxidation
- Three forms of ketone bodies can leave the liver: acetone, acetoacetate, and $\beta$-hydroxybutyrate.




Acetone




Acetoacetate




D-$\beta$-Hydroxybutyrate

**Unnumbered 17 p668**
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# Formation of Ketone Bodies: Generating Free CoA

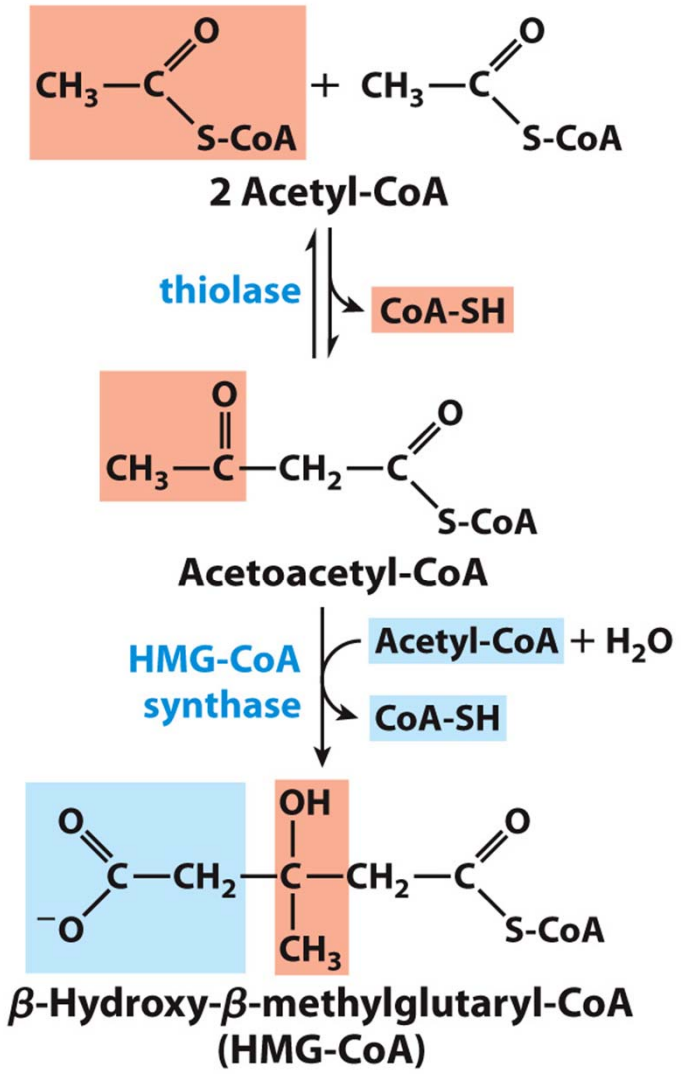



Figure 17-18 part 1
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- The first step is reverse of the last step in the $\beta$ oxidation: thiolase reaction joins two acetate units.
- A third acetyl-CoA is incorporated in the second step.
- Together, two CoA are freed from three acetyl-CoA.

# Formation of Ketone Bodies: Degradation of HMG-CoA

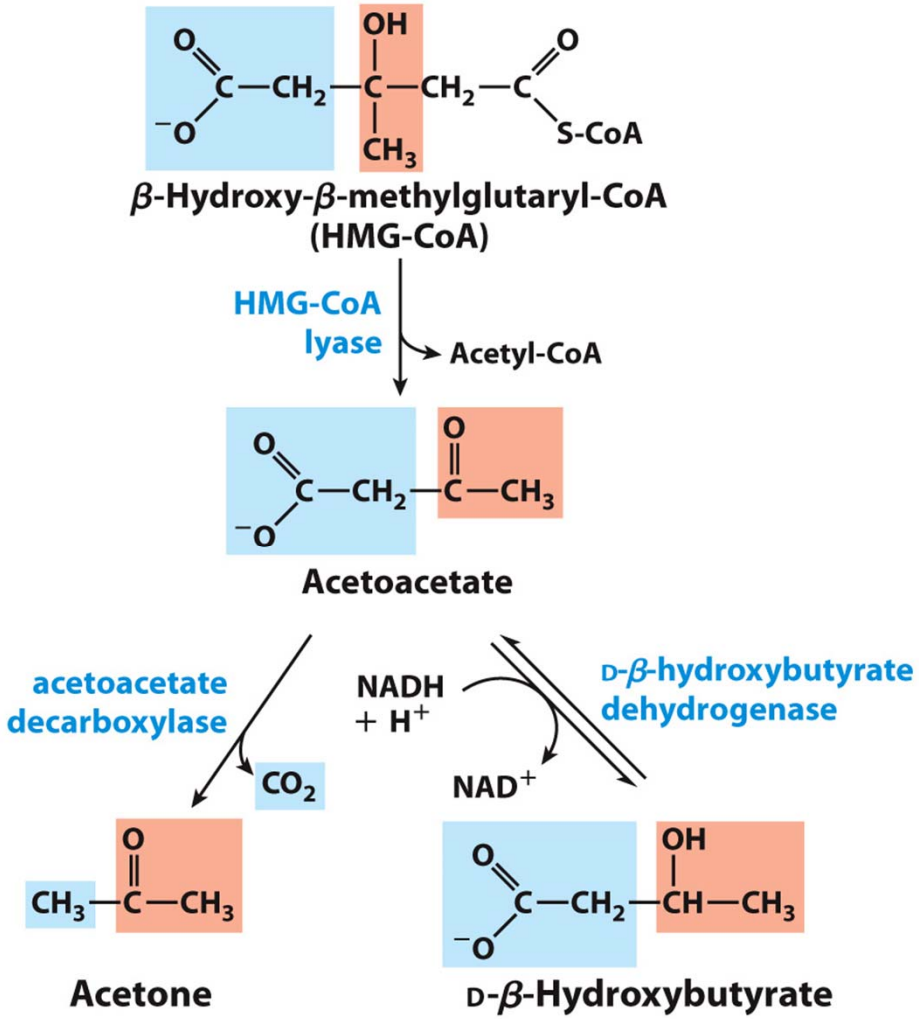



Figure 17-18 part 2
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- In order to traffic to other tissues, CoA must be removed. Acetone, acetoacetate, and β-hydroxybutyrate can then travel through the blood.
- Acetone is removed as a gas and exhaled, but acetoacetate and β-hydroxybutyrate can traffic to the brain for use in energy production.

# The Liver Is the Source of Ketone Bodies

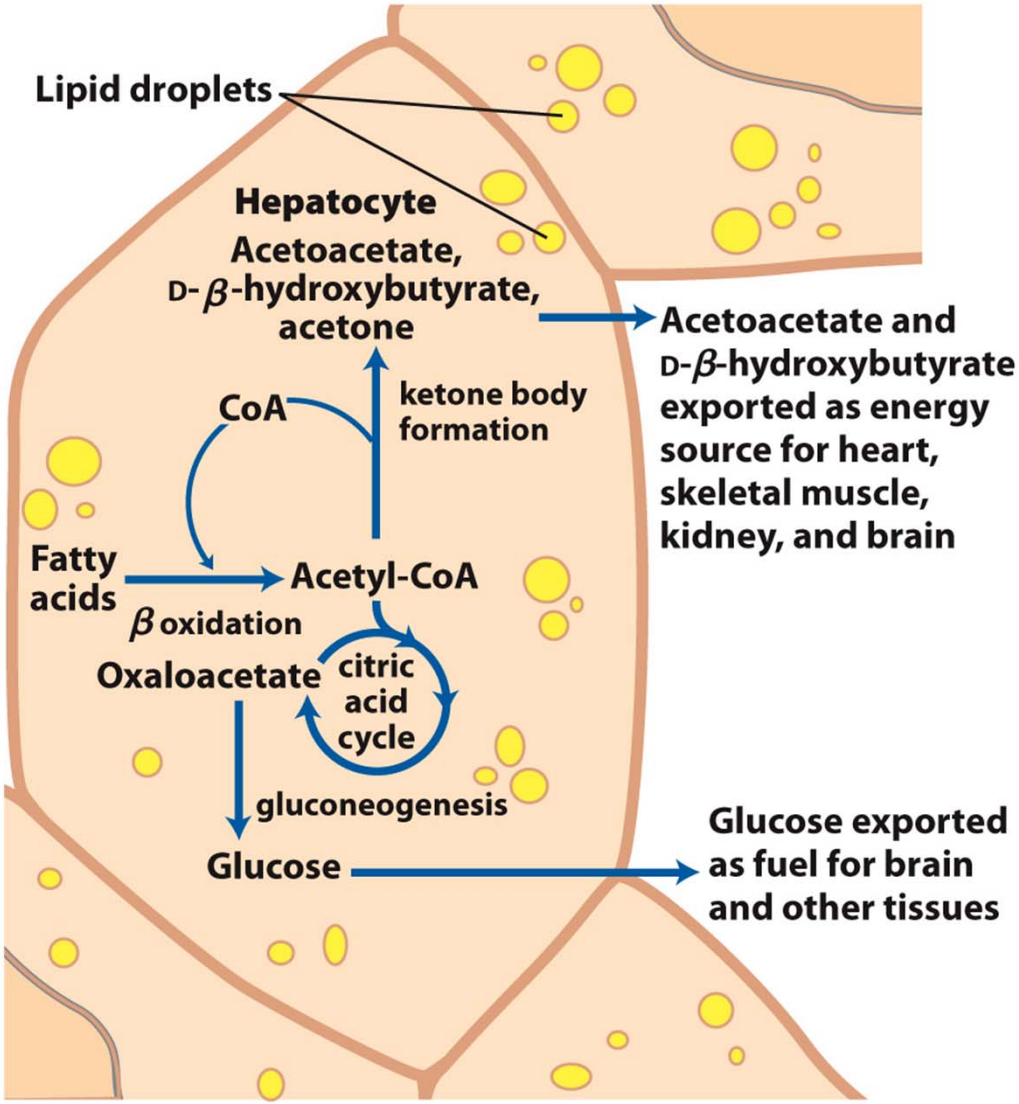



**Figure 17-20**
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# Ketone Bodies as Fuel

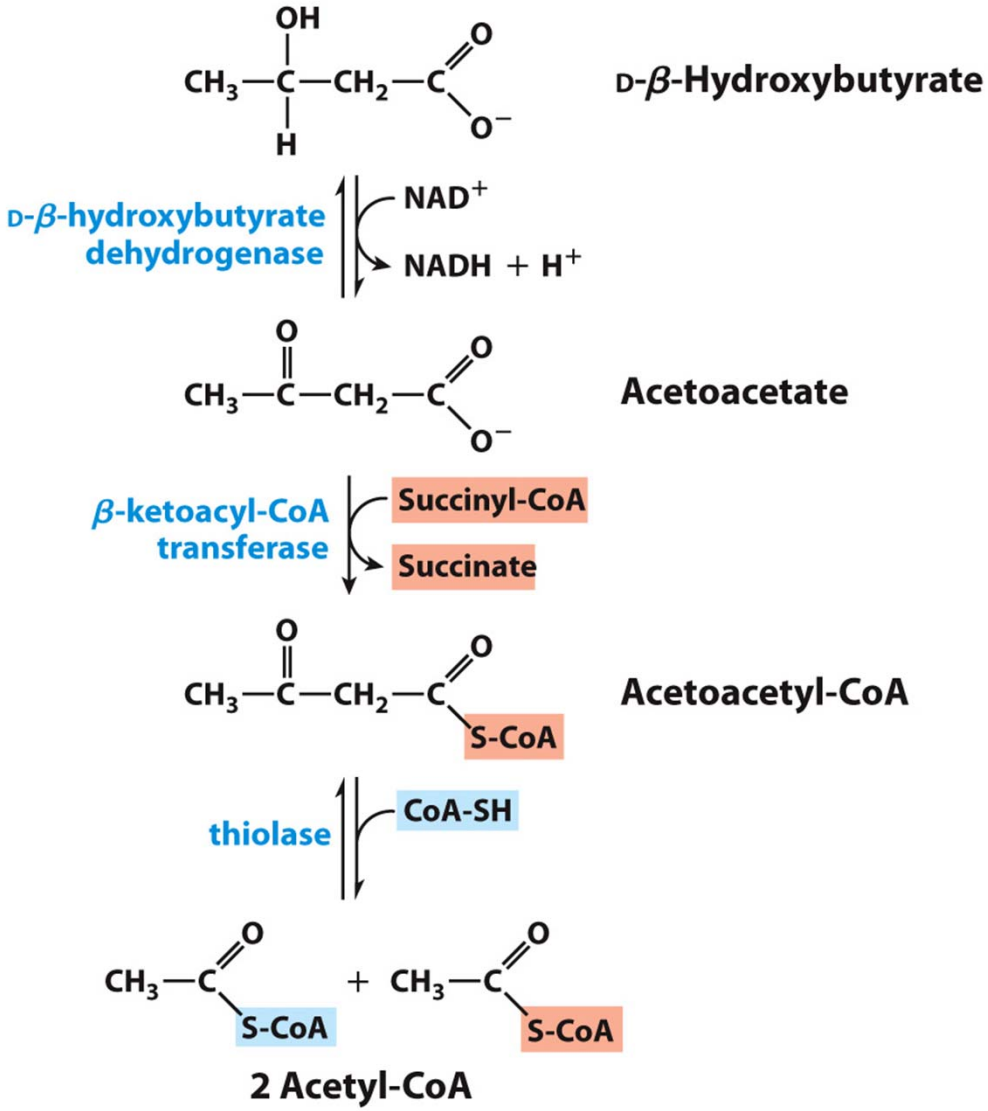



**Figure 17-19**
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table 17–2

## Ketone Body Accumulation in Diabetic Ketosis

| | Urinary excretion (mg/24 h) | Blood concentration (mg/100 mL) |
|---|---|---|
| Normal | ≤125 | <3 |
| Extreme ketosis (untreated diabetes) | 5,000 | 90 |